

Price Intelligence User Guide

1. Executive Summary

Price Intelligence is a historical procurement benchmarking feature for Budget Approval.

It helps approvers compare a proposed budget item price against QNH's own historical purchase behavior. Instead of reviewing only quantity, unit price, and total amount, approvers can now see whether the requested unit price is consistent with similar historical procurement records.

The feature was added because budget approval decisions often require context that was previously available only through manual investigation.

Example:

Budget Item	Quantity	Requested Unit Price
Office Chair	50	SAR 420

Without Price Intelligence, the approver must ask:

- Is SAR 420 reasonable?
- Has QNH purchased this item before?
- What did it usually cost?
- How much money is at risk if this price is high?

Price Intelligence answers these questions with historical procurement evidence.

The goal is procurement benchmarking and approval decision support. Procurement benchmarking is commonly used to compare proposed prices against historical purchasing behavior and market-aligned reference values. In Phase 1, this feature uses only QNH internal historical procurement data, not external market data.

How it helps key users:

User	How Price Intelligence Helps
CFO	Quickly identifies budget items with possible overspend risk and sees the financial impact.
Budget Approver	Reviews item-level price reasonableness without leaving the approval screen.
Procurement Team	Sees which historical item codes support the benchmark and can maintain mappings.
Auditor	Can trace benchmark results back to deterministic formulas and historical procurement records.
Developer	Can understand the business calculation model without reading source code.

Important principle:

Price Intelligence is not AI. It does not use language models, machine learning, embeddings, predictions, or external recommendation services. It uses deterministic calculations over approved source-of-truth mappings and historical PO records.

2. High-Level Data Flow

The high-level flow is:

```
Budget Item
|
v
Budget Type
|
v
PO Item Mappings
|
v
Mapped ITEM_CODE values
|
v
Historical PO Records
|
v
Price Intelligence Engine
|
v
Approval Indicators
```

Expanded flow:

```
HOD submits budget
|
v
Budget Approval page opens
|
v
System reads each budget item
|
v
System reads the item's Budget Type
|
v
System finds all active PO Item Mappings for that Budget Type
|
v
System collects all mapped ITEM_CODE values
|
v
System retrieves historical PO rows for those ITEM_CODE values
|
```

```
v
System calculates benchmark, statistics, variance, impact, and risk
|
v
CFO / Budget Approver sees table indicators and drawer details
```

Each step has a specific purpose:

Step	Meaning
Budget Item	The line item submitted for approval, such as Laptop or Office Chair.
Budget Type	The master item type behind the budget item.
PO Item Mappings	Maintained relationships between Budget Types and procurement ITEM_CODE values.
Mapped ITEM_CODEs	Stable item codes from Carware historical procurement data.
Historical PO Records	Actual QNH purchase invoice line records for mapped ITEM_CODEs.
Price Intelligence Engine	Deterministic calculation layer.
Approval Indicators	Benchmark, variance, potential overspend, status, and drawer details.

3. Understanding PO Item Mappings

A PO Item Mapping connects a budget type to a procurement item code.

Example:

```
Budget Type: Laptop

Laptop <-> ITEM_CODE 17819
Laptop <-> ITEM_CODE 17820
Laptop <-> ITEM_CODE 17821
```

Mappings exist because a single business item may appear under one or more procurement ITEM_CODE values in historical purchase data.

For example, Laptop may have multiple ITEM_CODE values because different laptop models, suppliers, or procurement classifications were used historically.

Mapping Source

Each mapping has a source:

Mapping Source	Meaning
MANUAL	Created manually by an administrator.

Mapping Source	Meaning
APPROVED_LINK	Learned automatically after an approved PO link confirmed a relationship.

MANUAL Mappings

Manual mappings are created by administrators when QNH already knows that a Budget Type corresponds to a PO ITEM_CODE.

Example:

```
Laptop <-> 17819
Source: MANUAL
```

Business meaning:

An administrator intentionally decided that historical PO records with ITEM_CODE 17819 should be considered relevant when benchmarking Laptop budget items.

APPROVED_LINK Mappings

Approved-link mappings are learned when a PO link is approved.

Example:

```
Approved PO Link:
Budget Item Type = Laptop
PO ITEM_CODE = 17820

System learns:
Laptop <-> 17820
Source: APPROVED_LINK
```

Business meaning:

An approver already confirmed a real-world relationship between that budget item type and the PO record. The system stores that relationship for future benchmarking and suggestions.

How Mappings Participate in Analysis

For benchmark calculations, MANUAL and APPROVED_LINK mappings are treated equally.

If Laptop has these mappings:

Budget Type	ITEM_CODE	Source
Laptop	17819	MANUAL
Laptop	17820	APPROVED_LINK

Budget Type	ITEM_CODE	Source
Laptop	17821	APPROVED_LINK

The engine uses all three active ITEM_CODE values:

```
17819
17820
17821
```

It does not use only manual mappings.

It does not use only learned mappings.

It does not use only the most recent mapping.

It does not use only one ITEM_CODE.

4. Historical Dataset Construction

Historical dataset construction is the most important part of the feature.

The engine must decide which historical PO records are relevant to the budget item being reviewed.

Step 1: Start With the Budget Item

Example budget item:

Field	Value
Budget Item	Office Chair
Budget Type ID	301
Quantity	50
Unit Price	SAR 420

The engine reads the Budget Type ID.

```
Office Chair budget item
|
v
Budget Type ID = 301
```

Step 2: Find Active Mappings

Mappings:

Budget Type	ITEM_CODE	Source	Active?
Office Chair	2102030060	MANUAL	Yes
Office Chair	2102030061	APPROVED_LINK	Yes
Office Chair	2102039999	MANUAL	No

The engine includes only active mappings.

Included:

```
2102030060
2102030061
```

Excluded:

```
2102039999
```

Reason:

Inactive mappings are disabled by administrators and should not influence benchmarks.

Step 3: Remove Duplicate ITEM_CODE Values

If duplicates exist:

Budget Type	ITEM_CODE	Source	Active?
Office Chair	2102030060	MANUAL	Yes
Office Chair	2102030060	APPROVED_LINK	Yes

The engine treats **2102030060** as one mapped code for benchmark construction.

This prevents the same historical PO rows from being counted twice.

Step 4: Retrieve Historical PO Records

The engine queries historical procurement records where:

```
BS_Purchase_Invoices_For_Budget.ITEM_CODE
matches one of the active mapped ITEM_CODE values
```

Example mappings:

```
2102030060
2102030061
```

Historical PO records:

PO Record	ITEM_CODE	Unit Cost	Included?
PO #1	2102030060	SAR 380	Yes
PO #2	2102030060	SAR 390	Yes
PO #3	2102030061	SAR 410	Yes
PO #4	2102039999	SAR 500	No

PO #4 is excluded because its ITEM_CODE is not active for Office Chair.

Step 5: Exclude Invalid Unit Costs

The engine includes only historical rows with:

```
UNIT_COST is not null
UNIT_COST > 0
```

Example:

PO Record	ITEM_CODE	Unit Cost	Included?
PO #1	2102030060	SAR 380	Yes
PO #2	2102030060	SAR 0	No
PO #3	2102030061	null	No

Zero or missing unit costs cannot support a price benchmark.

Step 6: Apply Evidence Window

The primary evidence window is the last 24 months.

Rule:

```
If at least one matching historical PO record exists in the last 24 months,
use only matching records from the last 24 months.
```

```
If no matching historical PO record exists in the last 24 months,
fall back to all available matching history.
```

Example A: Recent history exists

PO Record	Date	Unit Cost	Used?
PO #1	2026	SAR 390	Yes
PO #2	2025	SAR 395	Yes
PO #3	2020	SAR 310	No

The old 2020 record is excluded because recent evidence exists.

Example B: No recent history exists

PO Record	Date	Unit Cost	Used?
PO #1	2020	SAR 310	Yes
PO #2	2019	SAR 320	Yes

All available history is used because no recent evidence exists.

Summary of Dataset Rules

Condition	Behavior
Active mapping	Included.
Inactive mapping	Excluded.
Duplicate ITEM_CODE mapping	Counted once.
No mapping	No benchmark available.
Mapping exists but no PO history	No benchmark available, mapped codes still visible.
UNIT_COST is null or zero	Excluded.
Recent records exist	Use last 24 months only.
No recent records exist	Use all available matching history.

5. Every Metric Explained

Benchmark Price

Business Meaning

Benchmark Price is the main historical reference price used to judge whether the proposed budget unit price is reasonable.

It represents the typical historical unit cost for the mapped item codes.

Formula

Benchmark Price = Median Historical Unit Cost

The median is calculated from the scoped historical dataset.

Example

Historical unit costs:

350, 380, 390, 410, 900

Sorted:

350, 380, 390, 410, 900

Median:

390

Benchmark Price:

SAR 390

Interpretation

If the budget unit price is close to the Benchmark Price, the item is likely aligned with historical purchasing behavior.

If the budget unit price is much higher, the item may require review.

Average Price

Business Meaning

Average Price shows the arithmetic average of historical unit costs.

It is useful as a supporting statistic but is not the primary benchmark.

Formula

$$\text{Average Price} = \text{Sum of historical unit costs} / \text{Number of historical purchases}$$

Example

Historical unit costs:

350, 380, 390, 410, 900

Calculation:

$$\begin{aligned} & (350 + 380 + 390 + 410 + 900) / 5 \\ &= 2,430 / 5 \\ &= 486 \end{aligned}$$

Average Price:

SAR 486

Interpretation

The average can be pulled upward by outliers. In this example, SAR 900 makes the average much higher than the typical price.

Median Price

Business Meaning

Median Price is the middle value after historical unit costs are sorted.

It is more resistant to extreme prices than average.

Formula

Odd number of records:

$$\text{Median} = \text{middle value}$$

Even number of records:

Median = average of the two middle values

Example

Odd count:

350, 380, 390, 410, 900
Median = 390

Even count:

350, 380, 390, 410
Median = $(380 + 390) / 2 = 385$

Interpretation

Median is the current Benchmark Price.

Minimum Price

Business Meaning

Minimum Price is the lowest historical unit cost in the scoped dataset.

Formula

Minimum Price = Lowest historical UNIT_COST

Example

Historical unit costs:

350, 380, 390, 410

Minimum:

SAR 350

Interpretation

Minimum Price shows the lowest observed historical cost. It should not automatically be treated as the expected price because it may reflect an old deal, discount, promotion, or different item specification.

Maximum Price

Business Meaning

Maximum Price is the highest historical unit cost in the scoped dataset.

Formula

```
Maximum Price = Highest historical UNIT_COST
```

Example

Historical unit costs:

```
350, 380, 390, 410
```

Maximum:

```
SAR 410
```

Interpretation

Maximum Price shows the highest observed cost. If the requested budget price is above the maximum, that may require stronger justification.

Last Purchase Price

Business Meaning

Last Purchase Price is the unit cost from the most recent historical PO record in the scoped dataset.

Formula

```
Last Purchase Price = UNIT_COST from the most recent historical PO record
```

Example

Date	Unit Cost
01-Jan-2025	SAR 390
14-Jun-2026	SAR 405

Last Purchase Price:

SAR 405

Interpretation

Last Purchase Price shows the latest known procurement cost. It may reflect recent price movement, supplier change, or inflation.

Purchase Count

Business Meaning

Purchase Count is the number of historical PO records used in the benchmark calculation.

Formula

Purchase Count = Count of included historical PO records

Example

If 37 historical PO rows match active mapped ITEM_CODE values and pass the evidence window and unit cost rules:

Purchase Count = 37

Interpretation

Higher purchase count usually means stronger evidence. One record is useful, but less reliable than many records.

Supplier Count

Business Meaning

Supplier Count is the number of distinct suppliers represented in the historical dataset.

Formula

Supplier Count = Count of distinct non-empty supplier names

Example

Historical purchases:

PO	Supplier
PO #1	Supplier A
PO #2	Supplier A
PO #3	Supplier B
PO #4	Supplier C

Supplier Count:

3

Interpretation

More suppliers generally means the benchmark is less dependent on one vendor's pricing.

Benchmark Coverage

Business Meaning

Benchmark Coverage means whether a budget item has enough mapped historical procurement data to produce a benchmark.

This is shown indirectly through:

- Items Analyzed
- Missing Benchmarks
- No Benchmark Available status

Formula

For one item:

Covered = historical benchmark exists
Missing = no historical benchmark exists

For a budget:

```
Items Analyzed = count of items with benchmark  
Missing Benchmarks = count of items without benchmark
```

Example

Budget has 10 items:

```
7 items have benchmarks  
3 items do not
```

Coverage:

```
Items Analyzed = 7  
Missing Benchmarks = 3
```

Interpretation

Missing benchmarks show where mapping or historical data is incomplete.

Evidence Strength

Business Meaning

Evidence Strength explains how reliable the benchmark evidence is.

It is currently shown in the drawer and table as evidence information.

Formula

```
HIGH:  
  purchase_count >= 10  
  supplier_count >= 2  
  latest purchase age <= 24 months  
  
MEDIUM:  
  purchase_count >= 3  
  latest purchase age <= 36 months  
  
LOW:  
  purchase_count >= 1  
  
NONE:  
  purchase_count = 0
```

Example

```
Purchase Count = 12  
Supplier Count = 4  
Last Purchase = 6 months ago  
Evidence Strength = HIGH
```

Interpretation

HIGH evidence is more trustworthy than LOW evidence. LOW evidence should still be reviewed carefully.

Confidence Score

Confidence Score is not currently implemented as a numeric score.

The implemented substitute is Evidence Strength:

```
HIGH  
MEDIUM  
LOW  
NONE
```

This is intentionally simpler and easier to audit.

Historical Spend

Historical Spend is not currently shown as a main implemented metric.

The drawer shows recent historical purchase records with unit cost, quantity, and net amount data where available, but the feature does not currently display a formal Historical Spend KPI.

Variance Amount

Business Meaning

Variance Amount is the unit price difference between the budget request and the benchmark.

Formula

```
Variance Amount = Budget Unit Price - Benchmark Price
```

Example

```
Budget Unit Price = 420
Benchmark Price = 390
Variance Amount = 420 - 390 = 30
```

Interpretation

Positive variance means the budget price is above the benchmark.

Negative variance means the budget price is below the benchmark.

Variance Percent

Business Meaning

Variance Percent shows how far above or below benchmark the requested price is.

Formula

```
Variance % = ((Budget Unit Price - Benchmark Price) / Benchmark Price) * 100
```

Example

```
((420 - 390) / 390) * 100
= (30 / 390) * 100
= 7.69%
```

Displayed:

```
+7.7%
```

Potential Impact

Business Meaning

Potential Impact is the total financial difference between the budget request and the benchmark for the requested quantity.

Formula

```
Potential Impact = (Budget Unit Price - Benchmark Price) * Quantity
```

Example

$$\begin{aligned} & (420 - 390) * 50 \\ &= 30 * 50 \\ &= 1,500 \end{aligned}$$

Potential Impact:

SAR 1,500

Potential Overspend

Business Meaning

Potential Overspend is the positive financial risk amount. It ignores savings.

Formula

$$\text{Potential Overspend} = \max(0, \text{Potential Impact})$$

Example

$$\begin{aligned} \text{Potential Impact} &= \text{SAR } 1,500 \\ \text{Potential Overspend} &= \text{SAR } 1,500 \end{aligned}$$

If budget price is below benchmark:

$$\begin{aligned} \text{Potential Impact} &= -\text{SAR } 1,000 \\ \text{Potential Overspend} &= \text{SAR } 0 \end{aligned}$$

6. Benchmark Price Deep Dive

Benchmark Price is the most important price reference in the feature.

It is defined as:

$$\text{Benchmark Price} = \text{Median Historical Unit Cost}$$

The benchmark is not average price.

Why Median Was Chosen

Median is more resistant to outliers.

Example:

Historical prices:
350, 380, 390, 410, 900

Average:

$$(350 + 380 + 390 + 410 + 900) / 5 = 486$$

Median:

390

In this example, SAR 900 may be an unusual purchase, different specification, emergency purchase, or data outlier. Average is pulled upward. Median better reflects the normal historical price.

Example: Odd Number of Purchases

Historical prices:

380, 390, 410

Sorted:

380, 390, 410

Middle value:

390

Benchmark Price:

SAR 390

Example: Even Number of Purchases

Historical prices:

380, 390, 410, 420

Middle two values:

390 and 410

Median:

$(390 + 410) / 2 = 400$

Benchmark Price:

SAR 400

Example: Multiple ITEM_CODES

Budget Type:

Laptop

Mapped ITEM_CODES:

17819
17820
17821

Historical unit costs:

ITEM_CODE	Unit Cost
17819	3,900
17819	4,000
17820	4,100
17821	4,200

ITEM_CODE	Unit Cost
17821	4,500

Sorted:

3,900, 4,000, 4,100, 4,200, 4,500

Benchmark Price:

SAR 4,100

The benchmark uses all included records across all active mapped ITEM_CODES.

7. Variance Calculation

Variance shows how different the proposed budget unit price is from the Benchmark Price.

Formula:

$$\text{Variance \%} = ((\text{Budget Price} - \text{Benchmark Price}) / \text{Benchmark Price}) * 100$$

Example A: Budget Price Above Benchmark

Budget Price:

SAR 420

Benchmark:

SAR 390

Step 1:

```
Budget Price - Benchmark Price
= 420 - 390
= 30
```

Step 2:

$$\begin{aligned} & 30 / 390 \\ & = 0.0769 \end{aligned}$$

Step 3:

$$\begin{aligned} & 0.0769 * 100 \\ & = 7.69\% \end{aligned}$$

Displayed:

+7.7%

Interpretation:

The requested price is 7.7% above historical benchmark.

Example B: Budget Price Below Benchmark

Budget Price:

SAR 360

Benchmark:

SAR 390

Calculation:

$$\begin{aligned} & ((360 - 390) / 390) * 100 \\ & = (-30 / 390) * 100 \\ & = -7.69\% \end{aligned}$$

Displayed:

-7.7%

Interpretation:

The requested price is below historical benchmark. This may represent savings or a lower-specification item.

Example C: Budget Price Equals Benchmark

Budget Price:

SAR 390

Benchmark:

SAR 390

Calculation:

```
((390 - 390) / 390) * 100  
= 0%
```

Displayed:

0.0%

Interpretation:

The requested price exactly matches historical benchmark.

8. Potential Overspend

Potential Overspend is one of the most important CFO-facing values.

It translates a unit price difference into total budget risk.

Formula:

```
Potential Impact = (Budget Unit Price - Benchmark Price) * Quantity  
Potential Overspend = max(0, Potential Impact)
```

Why CFOs Care

A small unit price difference can become significant when multiplied by quantity.

Example:

```
Unit difference = SAR 30  
Quantity = 50
```

Potential Overspend = SAR 1,500

This helps finance leadership focus on material risk, not just percentage variance.

Example A: Positive Overspend

Budget Unit Price:

SAR 420

Benchmark:

SAR 390

Quantity:

50

Step 1:

$420 - 390 = 30$

Step 2:

$30 * 50 = 1,500$

Potential Overspend:

SAR 1,500

Interpretation:

If historical benchmark is a reasonable reference, approving this item at SAR 420 may create SAR 1,500 of avoidable budget exposure.

Example B: Price Below Benchmark

Budget Unit Price:

SAR 360

Benchmark:

SAR 390

Quantity:

50

Potential Impact:

$(360 - 390) * 50 = -1,500$

Potential Overspend:

SAR 0

Interpretation:

There is no overspend risk. The item is below historical benchmark. The approver may still verify that specifications are comparable.

Example C: Large Quantity Amplifies Risk

Budget Unit Price:

SAR 15 above benchmark

Quantity:

1,000

Potential Overspend:

$15 * 1,000 = \text{SAR } 15,000$

Interpretation:

Even moderate unit variance can matter when volume is high.

9. Risk Classification

Risk status is based on Variance Percent.

If no benchmark exists, status is No Benchmark Available.

Status	Color Meaning	Threshold	Business Meaning
Within Benchmark	Green	Variance <= 5%	Price is aligned with history.
Review Price	Yellow	> 5% and <= 15%	Price is moderately above benchmark.
Significant Variance	Orange	> 15% and <= 30%	Price is materially above benchmark.
High Overspend Risk	Red	> 30%	Price is far above benchmark.
No Benchmark Available	Gray	No benchmark	Not enough mapped history.

Within Benchmark

Threshold:

Variance <= 5%

Meaning:

The price is close to historical benchmark.

Recommended action:

Proceed with normal review. Check quantity and business justification.

Review Price

Threshold:

Variance > 5% and <= 15%

Meaning:

The price is above benchmark but not extreme.

Recommended action:

Ask whether inflation, specification changes, supplier change, or delivery requirements explain the variance.

Significant Variance

Threshold:

Variance > 15% and <= 30%

Meaning:

The requested price is materially above historical benchmark.

Recommended action:

Require explanation, compare recent purchases, and consider Procurement review.

High Overspend Risk

Threshold:

Variance > 30%

Meaning:

The requested price is significantly above historical benchmark and may represent a high financial risk.

Recommended action:

Escalate review. Confirm specifications, supplier conditions, and whether the budget should be revised.

No Benchmark Available

Threshold:

No historical benchmark exists

Meaning:

The system cannot determine price reasonableness from current mapping and historical data.

Recommended action:

Use manual review and consider creating or correcting PO item mappings.

10. Approval Guidance

Price Intelligence supports decision-making. It does not automatically approve or reject items.

If Item Is Green: Within Benchmark

Approver should think:

The requested price is consistent with historical purchases.

Recommended checks:

- Is the requested quantity justified?
- Is the department need valid?
- Is the budget total acceptable?

Example:

Benchmark = SAR 390
Budget Price = SAR 400
Variance = +2.6%
Status = Within Benchmark

Likely action:

Approve if business need and quantity are reasonable.

If Item Is Yellow: Review Price

Approver should think:

The price is somewhat above history. It may be reasonable, but should be checked.

Recommended checks:

- Has supplier pricing changed?
- Is the specification upgraded?
- Is this a different quality level?
- Is there inflation or urgent delivery?

Example:

Benchmark = SAR 390
Budget Price = SAR 420
Variance = +7.7%
Status = Review Price

Likely action:

Approve with justification, request clarification, or ask Procurement to confirm.

If Item Is Orange: Significant Variance

Approver should think:

The price is materially above historical benchmark.

Recommended checks:

- Require written justification.
- Review recent historical purchases.
- Check supplier count and evidence strength.
- Consider reducing budget price.

Example:

Benchmark = SAR 390
Budget Price = SAR 480
Variance = +23.1%
Status = Significant Variance

Likely action:

Return for revision unless a strong justification exists.

If Item Is Red: High Overspend Risk

Approver should think:

This item may create substantial avoidable overspend.

Recommended checks:

- Escalate to CFO or Procurement Manager.
- Verify item specification.
- Compare recent PO records.
- Require department explanation.
- Consider rejecting or returning the budget item.

Example:

Benchmark = SAR 390
Budget Price = SAR 550

```
Variance = +41.0%  
Status = High Overspend Risk
```

Likely action:

Return for revision unless there is strong evidence supporting the higher price.

If No Benchmark Is Available

Approver should think:

```
The system cannot provide historical price evidence for this item.
```

Recommended checks:

- Review manually.
- Ask Procurement whether historical ITEM_CODE mappings exist.
- Consider creating mappings for future approvals.

11. Drawer Explanation

The Details drawer explains how the benchmark was calculated.

Budget Information

Budget Type

Meaning:

The budget item type being reviewed, such as Laptop or Office Chair.

Source:

`BS_budget_items.type_id` joined to `BS_budget_types`.

Why it matters:

The Budget Type determines which PO item mappings are used.

Category

Meaning:

The budget category for the item.

Source:

`BS_budget_categories`.

Why it matters:

It provides business grouping and context.

Department

Meaning:

The department that submitted the budget.

Source:

`BS_budgets.department_id` joined to `BS_departments`.

Why it matters:

It identifies ownership of the request.

Financial Year

Meaning:

The financial year for the submitted budget.

Source:

`BS_budgets.financial_year_id` joined to `BS_financial_years`.

Why it matters:

Approval decisions are tied to a budget year.

Budget Quantity

Meaning:

The quantity requested by the department.

Source:

`BS_budget_items.quantity`.

Why it matters:

Quantity multiplies price variance into financial impact.

Budget Unit Price

Meaning:

The requested price per unit.

Source:

`BS_budget_items.unit_price`.

Why it matters:

It is compared against the Benchmark Price.

Budget Total

Meaning:

The total requested amount for the item.

Formula:

```
Budget Total = Quantity * Budget Unit Price
```

Source:

```
BS_budget_items.total_amount.
```

Why it matters:

It shows total budget exposure.

Benchmark Summary

Status

Meaning:

Risk classification based on variance from benchmark.

Source:

Calculated by Price Intelligence.

Why it matters:

It tells the approver how much review attention the item needs.

Benchmark Method

Meaning:

The method used to calculate benchmark.

Current value:

```
Median historical unit cost
```

Why it matters:

It explains why the benchmark may differ from average.

Historical Benchmark

Meaning:

The median historical unit cost.

Source:

Historical PO records for active mapped ITEM_CODE values.

Why it matters:

It is the main price reference.

Variance Amount

Formula:

$$\text{Budget Unit Price} - \text{Historical Benchmark}$$

Why it matters:

It shows unit-level difference.

Variance %

Formula:

$$((\text{Budget Unit Price} - \text{Historical Benchmark}) / \text{Historical Benchmark}) * 100$$

Why it matters:

It normalizes difference into percentage.

Potential Overspend

Formula:

$$\max(0, (\text{Budget Unit Price} - \text{Historical Benchmark}) * \text{Quantity})$$

Why it matters:

It shows the CFO-facing financial risk.

Historical Procurement Data

Historical Purchases

Meaning:

Number of historical PO records used.

Why it matters:

More records usually mean stronger evidence.

Suppliers

Meaning:

Number of distinct suppliers in the historical dataset.

Why it matters:

Multiple suppliers reduce dependency on one vendor's pricing.

Evidence Window

Meaning:

Which time range was used.

Possible values:

Value	Meaning
Last 24 months	Recent historical data exists and was used.
All available history	No recent records existed, so older history was used.

Mapped Item Codes

Meaning:

Active ITEM_CODE values used to gather historical records.

Why it matters:

It lets users verify whether the benchmark is based on the right procurement item codes.

Price Statistics

Median

Same as Historical Benchmark.

Average

Arithmetic mean of historical unit costs.

Minimum

Lowest historical unit cost.

Maximum

Highest historical unit cost.

Last Purchase

Most recent historical unit cost.

Last Purchase Date

Date of the most recent historical purchase in the scoped dataset.

Recent Historical Purchases

Meaning:

A limited table of recent PO records used as evidence.

Fields:

Field	Meaning
Date	PO record creation date.
Supplier	Supplier name from historical PO data.
Item Code	Procurement ITEM_CODE.
Description	Historical item description.
Quantity	Historical PO quantity.
Unit Cost	Historical unit cost.

Why it matters:

Approvers can inspect the actual records behind the benchmark.

12. Complete End-to-End Examples

Example 1: Office Chair

Budget item:

Field	Value
Budget Type	Office Chair
Quantity	50

Field	Value
Budget Unit Price	SAR 420

Mappings:

ITEM_CODE	Source	Active?
2102030060	MANUAL	Yes
2102030061	APPROVED_LINK	Yes

Historical PO records:

PO	ITEM_CODE	Date	Supplier	Unit Cost
PO #1	2102030060	2025	Supplier A	350
PO #2	2102030060	2025	Supplier A	380
PO #3	2102030061	2026	Supplier B	390
PO #4	2102030061	2026	Supplier C	410
PO #5	2102030060	2026	Supplier C	430

Sorted unit costs:

350, 380, 390, 410, 430

Benchmark:

Median = 390

Average:

$$\begin{aligned} &(350 + 380 + 390 + 410 + 430) / 5 \\ &= 1,960 / 5 \\ &= 392 \end{aligned}$$

Minimum:

350

Maximum:

430

Variance:

$$\begin{aligned} & ((420 - 390) / 390) * 100 \\ & = 7.69\% \end{aligned}$$

Potential Overspend:

$$\begin{aligned} & (420 - 390) * 50 \\ & = 30 * 50 \\ & = \text{SAR } 1,500 \end{aligned}$$

Risk:

Review Price

Reason:

Variance is greater than 5% and less than or equal to 15%.

Example 2: Laptop

Budget item:

Field	Value
Budget Type	Laptop
Quantity	20
Budget Unit Price	SAR 4,200

Mappings:

ITEM_CODE	Source	Active?
17819	MANUAL	Yes
17820	APPROVED_LINK	Yes
17821	APPROVED_LINK	Yes

Historical PO records:

PO	ITEM_CODE	Supplier	Unit Cost
PO #1	17819	Supplier A	3,800
PO #2	17819	Supplier A	3,900
PO #3	17820	Supplier B	4,000
PO #4	17821	Supplier C	4,100
PO #5	17821	Supplier C	4,500

Sorted unit costs:

3,800, 3,900, 4,000, 4,100, 4,500

Benchmark:

SAR 4,000

Average:

$$\begin{aligned} & (3,800 + 3,900 + 4,000 + 4,100 + 4,500) / 5 \\ &= 20,300 / 5 \\ &= 4,060 \end{aligned}$$

Variance:

$$\begin{aligned} & ((4,200 - 4,000) / 4,000) * 100 \\ &= 5\% \end{aligned}$$

Potential Overspend:

$$\begin{aligned} & (4,200 - 4,000) * 20 \\ &= 200 * 20 \\ &= SAR 4,000 \end{aligned}$$

Risk:

Within Benchmark

Reason:

Variance is equal to 5%, which remains within benchmark.

Example 3: Medical Equipment

Budget item:

Field	Value
Budget Type	Patient Monitor
Quantity	10
Budget Unit Price	SAR 18,000

Mappings:

ITEM_CODE	Source	Active?
PM-8840	MANUAL	Yes
PM-8841	APPROVED_LINK	Yes

Historical PO records:

PO	ITEM_CODE	Supplier	Unit Cost
PO #1	PM-8840	Supplier A	12,000
PO #2	PM-8840	Supplier B	12,500
PO #3	PM-8841	Supplier B	13,000
PO #4	PM-8841	Supplier C	13,500

Sorted unit costs:

12,000, 12,500, 13,000, 13,500

Benchmark:

$$(12,500 + 13,000) / 2$$

$$= 12,750$$

Variance:

$$((18,000 - 12,750) / 12,750) * 100$$

$$= 41.18\%$$

Potential Overspend:

```
(18,000 - 12,750) * 10  
= 5,250 * 10  
= SAR 52,500
```

Risk:

High Overspend Risk

Recommended action:

Escalate for review. Confirm whether the requested equipment has a different specification, warranty, supplier package, or clinical requirement.

13. Edge Cases

No Mappings

Condition:

No active `BS_PO_ITEM_MAPPINGS` rows exist for the Budget Type.

System displays:

No Benchmark Available

Benchmark:

null

Variance:

null

Potential Overspend:

SAR 0

Drawer message:

The drawer explains that no active PO item mappings exist.

No PO History

Condition:

Active mappings exist, but no matching historical PO records exist.

System displays:

No Benchmark Available

Drawer message:

Active mappings exist, but no purchase records were found for the mapped item codes.

One PO Only

Condition:

Only one historical PO record is available.

Behavior:

Median = that one unit cost
Average = that one unit cost
Minimum = that one unit cost
Maximum = that one unit cost
Purchase Count = 1

Evidence Strength:

LOW

Interpretation:

The benchmark exists but should be interpreted carefully.

One Supplier Only

Condition:

Historical records exist but all are from one supplier.

Behavior:

Supplier Count:

1

Evidence Strength cannot be HIGH because HIGH requires at least two suppliers.

Interpretation:

The benchmark may reflect one supplier's pricing rather than broader procurement behavior.

Extreme Outlier Prices

Condition:

Historical records include unusually high or low prices.

Example:

350, 380, 390, 410, 900

Behavior:

Median:

390

Average:

486

Interpretation:

Median protects the benchmark from being overly distorted by the outlier.

Disabled Mappings

Condition:

A mapping exists but `is_active = 0`.

Behavior:

The mapping is excluded from benchmark calculations.

Interpretation:

Administrators can disable mappings that are incorrect or no longer relevant.

Missing Benchmark

Condition:

No usable historical benchmark can be calculated.

Common causes:

- no active mappings
- active mappings but no PO history
- historical PO rows have zero or null unit cost

Behavior:

Status:

No Benchmark Available

Variance:

not calculated

Overspend:

SAR 0

14. Frequently Asked Questions

Why is benchmark different from average?

Benchmark uses median, not average.

Median is less affected by outliers.

Example:

350, 380, 390, 410, 900
Average = 486
Median = 390

The system uses SAR 390 as benchmark because it better represents the typical historical price.

Why is an item marked red?

An item is marked High Overspend Risk when:

Variance % > 30%

This means the requested unit price is more than 30% above historical benchmark.

Why is my budget price higher than historical price?

Possible reasons:

- inflation
- upgraded specification
- different supplier
- warranty or service package
- urgent delivery
- low historical price was unusual
- incorrect mapping

The approver should review the drawer details and request justification if needed.

Why is there no benchmark?

Common reasons:

- no active mapping exists for the Budget Type
- mappings exist but no historical PO records match
- historical records have missing or zero unit cost

How are mappings maintained?

Mappings are maintained through the PO Item Mapping Management page.

Administrators can:

- create mappings manually
- disable mappings
- review manual and learned mappings together

How does auto-learning work?

When a PO link is approved, the system learns the relationship between:

Budget Type
and
PO ITEM_CODE

If the mapping does not exist, it is created as:

```
APPROVED_LINK
```

If it already exists, the learning count and last learned date are updated.

Does Price Intelligence use only approved PO links?

No.

Approved PO links help create or update mappings, but benchmark calculations use:

```
BS_PO_ITEM_MAPPINGS  
BS_Purchase_Invoices_For_Budget
```

The benchmark is not calculated directly from `BS_PO_LINKS`.

Are manual and learned mappings weighted differently?

No.

For benchmark calculation, both are treated equally if active.

Does the system use external market data?

No.

Phase 1 uses QNH historical procurement data only.

15. Data Sources Reference

BS_budget_items

Role:

Stores submitted budget line items.

Used fields:

Field	Purpose
id	Identifies the budget item.
type_id	Links the budget item to its Budget Type.
quantity	Used in potential impact and overspend.
unit_price	Compared against benchmark.
total_amount	Displayed as budget total.
is_active	Ensures inactive items are excluded.

BS_budget_types

Role:

Defines master budget item types such as Laptop, Office Chair, Patient Monitor.

Used fields:

Field	Purpose
id	Used to match mappings.
name	Displayed as Budget Type.
category_id	Used for category display.
expense_type	Displayed for context.

BS_PO_ITEM_MAPPINGS

Role:

Source of truth for connecting Budget Types to procurement ITEM_CODE values.

Used fields:

Field	Purpose
budget_type_id	Connects mapping to Budget Type.
po_item_code	Historical procurement ITEM_CODE.
po_item_description	Description for administration and audit.
mapping_source	MANUAL or APPROVED_LINK.
is_active	Determines whether mapping participates in analysis.
learned_count	Supports mapping history, not benchmark weighting.
last_learned_at	Supports mapping history, not benchmark weighting.

BS_Purchase_Invoices_For_Budget

Role:

Historical procurement source table.

Used fields:

Field	Purpose
ID	Identifies historical PO line.
ITEM_CODE	Matched against mapped PO item codes.

Field	Purpose
ITEM_DESC	Displayed in drawer.
PARENT_ITEM_NAME	Displayed in drawer where available.
SUPPLIER_NAME_EN	Used for supplier count and drawer display.
QTY	Displayed in recent purchases.
UNIT_COST	Main input to all price calculations.
NET_AMOUNT	Displayed where available.
CREATED_AT	Used for evidence window and last purchase.

BS_PO_LINKS

Role:

Audit history and learning source.

Price Intelligence does not calculate benchmarks directly from this table.

It contributes indirectly because approved PO links can create or update mappings in [BS_PO_ITEM_MAPPINGS](#).

16. Formula Reference Sheet

Benchmark Price

```
Benchmark Price = Median Historical Unit Cost
```

Median: Odd Number of Records

```
Median = Middle value after sorting unit costs ascending
```

Median: Even Number of Records

```
Median = Average of the two middle values
```

Average Price

```
Average Price = Sum(Historical Unit Costs) / Purchase Count
```

Minimum Price

$$\text{Minimum Price} = \text{MIN}(\text{Historical Unit Cost})$$

Maximum Price

$$\text{Maximum Price} = \text{MAX}(\text{Historical Unit Cost})$$

Last Purchase Price

$$\text{Last Purchase Price} = \text{Unit Cost of most recent included PO record}$$

Purchase Count

$$\text{Purchase Count} = \text{Number of included historical PO records}$$

Supplier Count

$$\text{Supplier Count} = \text{Number of distinct non-empty supplier names}$$

Variance Amount

$$\text{Variance Amount} = \text{Budget Unit Price} - \text{Benchmark Price}$$

Variance Percent

$$\text{Variance \%} = ((\text{Budget Unit Price} - \text{Benchmark Price}) / \text{Benchmark Price}) * 100$$

Potential Impact

$$\text{Potential Impact} = (\text{Budget Unit Price} - \text{Benchmark Price}) * \text{Quantity}$$

Potential Overspend

Potential Overspend = $\max(0, \text{Potential Impact})$

Items Analyzed

Items Analyzed = Count of budget items with an available benchmark

Missing Benchmarks

Missing Benchmarks = Count of budget items with no available benchmark

High Risk Items

High Risk Items = Count of items where Status = High Overspend Risk

Evidence Strength

HIGH =
Purchase Count ≥ 10
and Supplier Count ≥ 2
and Last Purchase Age ≤ 24 months

MEDIUM =
Purchase Count ≥ 3
and Last Purchase Age ≤ 36 months

LOW =
Purchase Count ≥ 1

NONE =
Purchase Count = 0

Risk Status

No Benchmark Available:
no benchmark exists

Within Benchmark:
Variance $\leq 5\%$

Review Price:

Variance > 5% and <= 15%

Significant Variance:

Variance > 15% and <= 30%

High Overspend Risk:

Variance > 30%

Historical Dataset Rule

Dataset =

Historical PO records where:

ITEM_CODE is in active mappings for the Budget Type
and UNIT_COST is not null
and UNIT_COST > 0

If recent records exist in the last 24 months:

use only last 24 months

Otherwise:

use all available matching history